

$$(u_g)_x = \frac{1}{y} \int_{-y}^y \underline{g_x(x+t)} dt \quad \checkmark$$

$$(u_g)_{xx} = \frac{1}{y} \int_{-y}^y g_{xx}(x+t) dt$$

$$(u_g)_y = -\frac{1}{y^2} \int_{-y}^y g(x+t) dt + \frac{1}{y} [g(x+y) + g(x-y)]$$

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$$F(x) = \int_{\psi(x)}^{\psi(x)} f(t) dt$$

$$\frac{d}{dx} F(x) = \underbrace{f(\psi(x))}_{\psi'(x)} \underbrace{\psi'(x)}_{\psi'(x)} - \underbrace{f(\psi(x))}_{\psi'(x)}$$

↓

FTOC

$$+ \int_{\psi(x)}^{\psi(x)} \underbrace{\frac{\partial}{\partial x} f(t)}_{=0} dt$$

$$F(x) = \int_{\psi(x)}^{\psi(x)} f(x, t) dt$$

$$\frac{d}{dx} F(x) = f(x, \psi(x)) \psi'(x) - f(x, \psi(x)) \psi'(x) + \int_{\psi(x)}^{\psi(x)} \frac{\partial}{\partial x} f(x, t) dt$$

$$u_g(x, y) = \frac{1}{y} \int_{-y}^y g(x+t) dt$$

$$\frac{\partial}{\partial x} u_g(x, y) = \frac{1}{y} \left[ g(x+y) \frac{\partial}{\partial x}(y) - g(x-y) \frac{\partial}{\partial x}(-y) + \int_{-y}^y \frac{\partial}{\partial x} g(x+t) dt \right]$$

$$= \frac{1}{y} \int_{-y}^y \frac{\partial}{\partial x} g(x+t) dt$$

$$= \frac{1}{y} \int_{-y}^y g_x(x+t) dt$$

$$(u_g)_{xx} = \left( \frac{1}{y} \int_{-y}^y g(x+t) dt \right)$$

$$u_g = \frac{1}{y} \int_{-y}^y g(x+t) dt$$

$$(u_g)_y = -\frac{1}{y^2} \int_{-y}^y g(x+t) dt$$

$$+ \frac{1}{y} \left[ g(x+y) \frac{\partial}{\partial y}(y) \right.$$

$$\left. - g(x-y) \frac{\partial}{\partial y}(-y) \right.$$

$$\left. + \int_{-y}^y \frac{\partial}{\partial y} g(x+t) dt \right]$$

$$= -\frac{1}{y^2} \int_{-y}^y g(x+t) dt$$

$$+ \frac{1}{y} [g(x+y) + g(x-y)]$$

$$(u_g)_{yy} = \frac{2}{y^3} \int_{-y}^y g(x+t) dt$$

$$- \frac{1}{y^2} [g(x+y) + g(x-y)]$$

$$-\frac{1}{y^2} [g(x+y) + g(x-y)] \\ + \frac{1}{y} [g_y(x+y) + g_y(x-y)]$$

$$\sqrt{3} \ln(\sqrt{3}) \\ = \frac{\sqrt{3}}{2} \ln(3)$$

$$\frac{dy}{dx} = f(x, y)$$

$$\frac{dy}{dx} = -\frac{1}{f(x, y)}$$

$$\frac{dy}{dx} = -\frac{1}{f(x)}$$

$$\checkmark \frac{dy}{dx} = f(x, y)$$

$$\frac{dy}{dx} \times \frac{dx}{dy} = -1$$

$$\checkmark \frac{dy}{dx} = g(x, y)$$

$$fg = -1$$

$$F(x,y) = 0, h(x,y) = 0$$

orthogonal

$$\left( \frac{dy}{dx} \right)_F \times \left( \frac{dy}{dx} \right)_h = -1$$

$10 = 2 \cdot 5$

$$m_1 m_2 = -1$$

$$|U(10)| = \varphi(10)$$

$$= 10 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right)$$

$$= 10 \times \frac{1}{2} \times \frac{4}{5}$$

$$= 4$$

$$|U(10)| = \varphi(10) = 4.$$

$$|U(5)|$$

$$|U(n)| = \varphi(n)$$

$$n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

$$n = 2^1 3^1 5^1 \dots$$

$$100 = 2^2 5^2$$

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right)$$

$$\varphi(\underline{100}) = 100 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right)$$

$$= 100 \times \frac{1}{2} \times \frac{4}{5}$$

$$= \textcircled{40}$$

$$S_n \neq \mathbb{Z}_{n!}$$

$$\textcircled{n!} \quad \textcircled{n!}$$

$$8 = 2^3$$

$$|U(5)| = 5 \left(1 - \frac{1}{5}\right)$$

$$= 5 \times \frac{4}{5} = 4$$

$$|U(8)| = 8 \times \left(1 - \frac{1}{2}\right) = 8 \times \frac{1}{2} = 4$$

$$|U(10)| = 10 \times \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) \quad \textcircled{f: \textcircled{4_1} \rightarrow \textcircled{4_2}}$$

↓  
lin. 1-1, onto

$$= 10 \times \frac{1}{2} \times \frac{4}{5} = 4$$

$$12 = 2^2 \times 3$$

$$|U(12)| = 12 \times \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right)$$

$$= 12 \times \frac{1}{2} \times \frac{2}{3} = 4$$

$$|U(8)| = 4 = |U(10)|$$

$$\textcircled{4} \rightarrow \mathbb{Z}_4$$

$$\textcircled{4} \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$$

$$2^2$$

$$\textcircled{p^2}$$

Note:  $U(n)$  is cyclic iff  $n=1, 2, 4, p^k$  or

$2p^k$  :  $\rightarrow$

$2p^k$   
 $\downarrow$   
 $p \neq 2$

$n=5$ ,  $n=8$   
 $= 2$

$n=10 = 2 \times 5$

$n=12 = 2^2 \times 3$

$U(10)$

$\neq$

$U(12) \times$

$U(8)$  is not cyc.  $n=8=2^3$

$U(10) \rightarrow$  cyclic

$U(8) \neq U(10)$

$42 \mid |S_{13}| = 13! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10 \times 11 \times 12 \times 13$

$11$

$2 \times 3 \times 7$

$35$   
 $11$   
 $5 \times 7$

$$3^3 \mid 13!$$

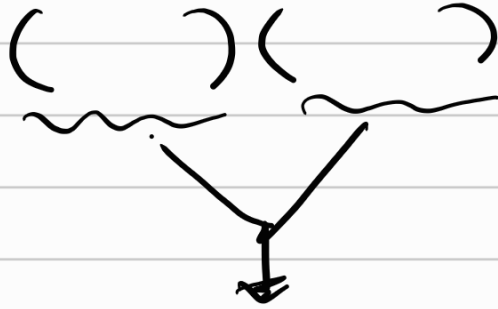
$$\textcircled{S_n}$$

$$\textcircled{(1\ 2\ 3\ \dots\ n)}$$

$$60$$

$$11$$

$$\underline{2^2 + 3 \times 5}$$



$$\textcircled{42 = 2 \times 3 \times 7}$$

$$\underbrace{(\dots)}_{\sim} \underbrace{(\dots)}_{\sim} \underbrace{(\dots)}_{\sim}$$

$$< \textcircled{13}$$

$$35 = 5 \times 7$$

$$\underbrace{(\overset{\vee}{1}\ \overset{\vee}{2})}_{2} \underbrace{(\overset{\vee}{3}\ \overset{\vee}{4}\ \overset{\vee}{5})}_{3} \underbrace{(\overset{\vee}{6}\ \overset{\vee}{7}\ 8\ 9\ 10\ 11\ 12)}_{7}$$

$$\checkmark \underbrace{(\underbrace{1\ 2\ 3\ 4\ 5}_{5})}_{2} \underbrace{(\underbrace{6\ 7\ 8\ 9\ 10\ 11\ 12}_{7})}_{7}$$



$$3 + 3 + \textcircled{3} \quad \checkmark \quad \checkmark$$

$$\underbrace{(\dots) (\dots) (\dots)}_{\text{LCM}(3,3,3) = \textcircled{3}}$$

$p^k$

$$60 = 2^2 \times 3 \times \textcircled{5}$$

$$< 13$$

$$(1 \ 2 \ 3 \ 4) \ (5 \ 6 \ 7)$$

$$\underbrace{\hspace{10em}}_{\textcircled{2}} \quad \underbrace{(8 \ 9 \ 10 \ 11 \ 12)}_5$$

$$m = \overset{S_n}{\underbrace{p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}}_{\substack{p_1^{k_1} \rightarrow (\dots) \\ p_2^{k_2} \rightarrow (\dots) \\ p_n^{k_n} \rightarrow (\dots)}}} \quad ? < n$$

In  $S_n$ ,  $m$  order element exists  
iff (if  $m = \underbrace{p_1^{k_1}}_{\text{circled}} p_2^{k_2} \dots p_r^{k_r}$ , then

$$\underbrace{p_1^{k_1}}_{\text{circled}} + \underbrace{p_2^{k_2}}_{\text{underlined}} + \dots + \underbrace{p_r^{k_r}}_{\text{underlined}} < n)$$

↙  $n=13$

$$m=42 = 2 \times 3 \times 7$$

$$2 + 3 + 7 = 12 < 13 \quad \checkmark$$

$$m=35 = 5 \times 7$$

$$5 + 7 = 12 < 13$$

$$m=3^3=27$$

$$\underbrace{3}_{\text{circled}} = 27 > 13$$

$$\checkmark \quad m=60 = 2^2 \times 3 \times 5$$

$$\underline{2} + \underline{3} + \underline{5} = 12 < 13$$

